

CHAPTER 1: INTRODUCTION

1.1 Background

Agriculture plays a significant role in the economic development of many countries. Efficient irrigation is essential for maintaining healthy crop growth and maximizing agricultural productivity. Traditional irrigation methods often result in excessive water consumption due to fixed watering schedules and manual monitoring. This leads to water wastage, increased labor costs, and reduced efficiency.

Recent advancements in the Internet of Things (IoT) have enabled the development of intelligent agricultural systems capable of monitoring environmental conditions and automating irrigation processes. By integrating sensors, microcontrollers, and communication technologies, smart irrigation systems can provide real-time data and make automated decisions based on soil conditions.

The ESP32 microcontroller is widely used in IoT applications because of its processing power, low cost, integrated Wi-Fi, and extensive peripheral support. In this project, ESP32 acts as the central controller responsible for acquiring sensor data, processing information, displaying results, and controlling irrigation.

1.2 Need for Smart Irrigation

The increasing demand for food production and the limited availability of water resources require efficient water management systems.

Major problems with traditional irrigation include:

- Over-irrigation
- Under-irrigation
- Water wastage
- High labor requirements
- Lack of environmental monitoring

Smart irrigation systems address these challenges by supplying water only when necessary.

1.3 Project Overview

The IoT Smart Irrigation and Plant Monitoring System is designed to monitor soil moisture, temperature, and humidity continuously.

The system consists of:

- ESP32 Development Board
- DHT11 Temperature and Humidity Sensor
- Soil Moisture Sensor
- OLED Display
- Relay Module
- Water Pump

The soil moisture sensor detects the moisture content present in the soil. The DHT11 sensor measures temperature and humidity. The ESP32 processes sensor readings and displays the results on an OLED screen.

Whenever soil moisture falls below a predefined threshold, the ESP32 activates a relay module which controls a water pump. The pump irrigates the plant automatically until sufficient moisture is detected.

1.4 Objectives

The main objectives of this project are:

1. To monitor soil moisture continuously.
 2. To measure temperature and humidity.
 3. To display sensor data in real time.
 4. To automate irrigation using relay control.
 5. To reduce water wastage.
 6. To demonstrate IoT concepts in agriculture.
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1.5 Scope of the Project

The scope of this project includes:

- Small-scale agricultural applications

- Home gardening systems
- Greenhouse monitoring
- Indoor plant care
- Educational and research purposes

The system can be expanded in the future through Wi-Fi connectivity and cloud-based monitoring.